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| 16 | <p>Unpolarised light first passes through a polarising filter with its axis vertical. Its intensity is reduced to <math>175 \text{ W m}^{-2}</math>. The polarised light then passes through a second polarising filter. It emerges from the second filter with an intensity of <math>131 \text{ W m}^{-2}</math>.</p> <p>Which is the angle from the vertical of the axis of the second polarising filter?</p> |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|   |            |   |            |   |            |   |            |
|---|------------|---|------------|---|------------|---|------------|
| A | $30^\circ$ | B | $42^\circ$ | C | $48^\circ$ | D | $60^\circ$ |
|---|------------|---|------------|---|------------|---|------------|

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|  | <p><b>Answer: A</b></p> <p>Let <math>I_0</math> be intensity of the polarised light after passing through the first polarising filter = <math>175 \text{ W m}^{-2}</math>, and<br/> <math>I_t</math> be intensity of the polarised light after passing through the second polarising filter = <math>131 \text{ W m}^{-2}</math>.</p> <p>Using Malus' Law,<br/> <math>I_t = I_0 \cos^2 \theta</math><br/> <math>131 = 175 \cos^2 \theta</math><br/> <math>\theta = 30^\circ</math></p> |
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| 17 | A standing wave is set up on a rope of length 1.0 m fixed at both ends. |
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